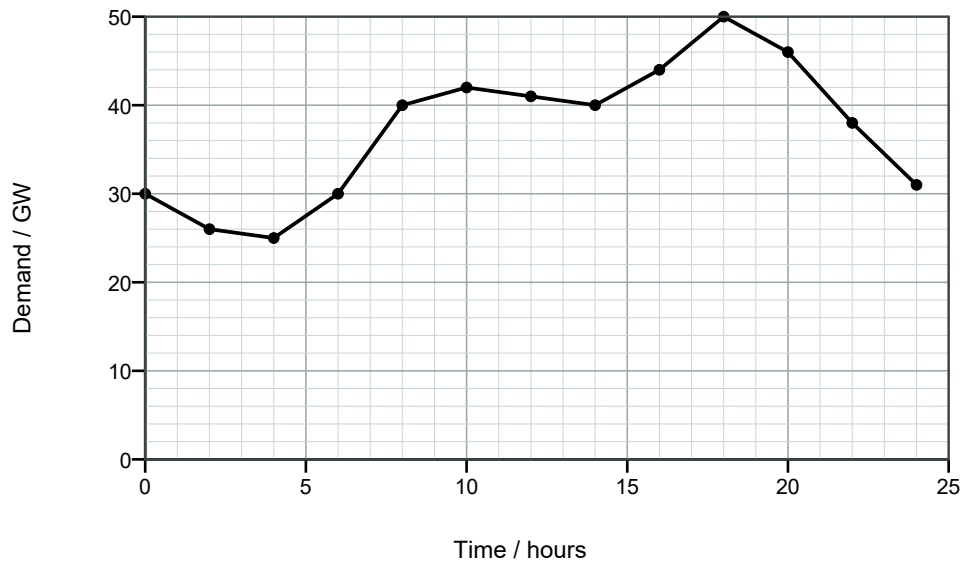
Questions by topic: Fluctuations in supply/demand & energy storage - Set 1 (60 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 1 . 1 Describe the pattern of demand shown in Figure 1.

[2 marks]

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0 1 . 2 Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

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0 1 . 3 Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

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0 1 . 4 Give **two** causes of fluctuations in energy **supply**.

[2 marks]

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0 2 A pumped-storage HEP station has an upper reservoir 500 m above its turbines. It can hold 7 million m^3 of water. The density of water is 1000 kg m^{-3} and g = 9.8 N kg^{-1}.

0 2 . 1 Describe how pumped-storage HEP is used for peak shaving.

[3 marks]

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0 2 . 2 Calculate the maximum gravitational potential energy stored when the upper reservoir is full. Give your answer in standard form.

[3 marks]

Answer ☐ J

0 2 . 3 The station returns 75% of the electricity used for pumping. Explain why the process is still worthwhile despite this loss.

[2 marks]

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0 2 . 4 Give **two** locational requirements of a pumped-storage station.

[2 marks]

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0 3 . 1 Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

0 3 . 2 Compare **three** methods of storing hydrogen, giving a disadvantage of each.

[3 marks]

0 3 . 3 Explain how a fuel cell differs from a rechargeable battery.

[2 marks]

0 3 . 4 Explain what is meant by the 'hydrogen economy'.

[1 mark]

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0 4 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg^{-1}	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

0 4 . 1 Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

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0 4 . 2 Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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0 4 . 3 Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

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0 4 . 4 Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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0 5 . 1 Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

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0 5 . 2 Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

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0 5 . 3 Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

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0 5 . 4 Explain how high thermal mass building materials reduce energy use.

[2 marks]

0 5 . 5 Describe **one** other energy storage technology.

[2 marks]

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0 6 . 1 Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

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0 6 . 2 State what is meant by **peak shaving**.

[1 mark]

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END OF QUESTIONS

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